

NON-MF GROUPS

SAUERS

We give a negative answer to the longstanding question whether every countable group is MF in the operator norm sense. Let $L \leq G$ and $K \trianglelefteq G$ have property (T). If K is contained in the compression–centralizer defect of L , then K lies in the MF radical of G . For $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$, this defect equals H . The group H is finitely generated, simple, has property (T), and has trivial image under every homomorphism to an MF group. Consequently, $C_r^*(H)$ is a separable stably finite C^* -algebra that is not MF. We also construct a finitely presented torsion-free property-(T) group with full MF radical; every nontrivial quotient of it again has full MF radical.

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1. INTRODUCTION

Not every countable group is MF. The counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

Khanh–Thanh show that H is isomorphic to the unit group of the binary Leavitt algebra [K–T, Proposition 4.2 and Corollary 4.4], which OpenAI used to construct the first nonsofic group [OAI, Chapter 3].

A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

We use this operator norm convention throughout; the separation may equivalently be required along the full sequence with a constant independent of g [Kor, Propositions 2 and 7]. The strong convergence convention additionally requires the matrix models to recover the norms of the left regular representation [GKM, Sch], so a group that is not MF in the present sense is not MF in that sense either.

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Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here $\prod_n M_{d_n}(\mathbb{C})$ is the C^* -algebra of bounded sequences, as every product of C^* -algebras below, and $\bigoplus_n M_{d_n}(\mathbb{C})$ is its ideal of sequences (x_n) with $\|x_n\| \rightarrow 0$.

A separable C^* -algebra is MF if it embeds in a norm matrix corona. For a countable group G , the algebra $C_r^*(G)$ is separable and its faithful canonical trace makes it stably finite. The group H above gives a non-MF algebra of this form.

For a countable group G , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. By Proposition 2.1, G is MF if and only if $\text{Rad}_{\text{MF}}(G) = 1$, while $\text{Rad}_{\text{MF}}(G) = G$ implies that every homomorphism from G to an MF group is trivial.

For a subgroup $L \leq G$, put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

Theorem A (one-sided compression criterion). *Let G be countable and let $L \leq G$ have property (T). If $K \trianglelefteq G$ has property (T) and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) and $\mathfrak{D}_G(L) = G$, then $\text{Rad}_{\text{MF}}(G) = G$.

Every operator norm asymptotic representation is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm; in the notation of Section 2,

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Conjugation by the compressor moves the Kazhdan projection of L below itself, and stable finiteness of the corona forces equality. For a corona representation, the complement of the Kazhdan projection is a G -invariant corner. The Kazhdan inequality forces this corner to vanish when $K \leq \mathfrak{D}_G(L)$.

Many Kazhdan groups are residually finite and so MF.

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group). *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Every homomorphism from H to an MF group is trivial. So H is non-MF, and $C_r^(H)$ is separable and stably finite but not MF.*

Theorem C (a torsion-free group with full MF radical). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property (T) and*

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

Every nontrivial quotient of Q also equals its own MF radical; in particular, no nontrivial quotient of Q is MF.

The construction combines Fournier-Facio's torsion-free property-(T) group [FF, §2] with Hull's small cancellation theorem [Hu] so that the compression defect normally generates Q .

Relation to prior work. Blackadar and Kirchberg developed the notions of MF and NF algebras and proved that a separable C^* -algebra is NF if and only if it is nuclear and MF [BK]. Carrión-Dadarlat-Eckhardt introduced the group property used here [CDE]. The negative solution of the Connes embedding problem [JNVWY] implies the existence of separable stably finite C^* -algebras that are not MF [GH, Proposition 6.1 and Remark 6.2]; Theorem B gives one among reduced group C^* -algebras. Nonapproximability was known for the unnormalized Schatten p -norms when $1 < p < \infty$ [DGLT, LO], and Bachner-Dogon-Lubotzky settled $p = 1$ [BDL]. The case $p = \infty$ remained open. For groups of Deligne type, operator-Hilbert-Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]. Here the input is the compression relation: conjugation by the compressor preserves the normalized Hilbert-Schmidt asymptotic commutant.

Leavitt introduced the algebras $L_K(1, n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then implies that H is simple [Pre]. Property (T) follows from Ershov-Jaikin-Zapirain [EJZ, Theorem 1.1].

2. ONE-SIDED COMPRESSION AND THE MF RADICAL

2.1. The MF radical. The image of a corona homomorphism from G is a countable subgroup of a corona unitary group, so MF; conversely, every countable MF group embeds in the unitary group of a norm matrix corona. Thus $\text{Rad}_{\text{MF}}(G)$ is also the intersection of the kernels of all homomorphisms from G to MF groups.

Proposition 2.1 (basic properties of the MF radical). *For every countable group G , the subgroup $\text{Rad}_{\text{MF}}(G)$ is fully invariant, the quotient $G/\text{Rad}_{\text{MF}}(G)$ is MF, and G is MF if and only if $\text{Rad}_{\text{MF}}(G)$ is trivial.*

Proof. An intersection of kernels is normal. If $\alpha: G \rightarrow G$ is an endomorphism and $x \in \text{Rad}_{\text{MF}}(G)$, then every corona homomorphism π satisfies $\pi(\alpha(x)) = (\pi \circ \alpha)(x) = 1$. Thus the radical is fully invariant.

Put $R = \text{Rad}_{\text{MF}}(G)$. If $1 \neq x \in G/R$ and g represents x , then $g \notin R$, so a homomorphism from G to an MF group separates g from 1. It kills R and induces a homomorphism separating x . So G/R is residually MF, and MF by Korchagin [Kor, Proposition 6]; this also covers the trivial quotient.

Finally, a corona embedding gives $\text{Rad}_{\text{MF}}(G) = 1$ when G is MF, and the quotient result gives the converse. $\square$

2.2. Kazhdan transport in normalized Hilbert–Schmidt norm. For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \rightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, and this norm is unitarily invariant; so $\|V_n(gh) - 1\|_2 \leq \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1)$ and $\|V_n(ghg^{-1}) - 1\|_2 = \|V_n(h) - 1\|_2 + o(1)$, and $K_2(V)$ is a normal subgroup of G . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator norm asymptotic representations of G .

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \rightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\text{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.2. *For every sequence of positive integers (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. Suppose that $v^*v = 1$ in A , and let (x_n) be a bounded lift of v . Then $\|x_n^*x_n - 1\| \rightarrow 0$. For all sufficiently large n , the matrix $x_n^*x_n$ is invertible, and

$$u_n = x_n(x_n^*x_n)^{-1/2}$$

is unitary. Moreover, $\|u_n - x_n\| \rightarrow 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v . So $vv^* = 1$, and A is finite. For every $k \geq 1$, apply this construction with matrix sizes km_n to see that $\prod_n M_{km_n}(\mathbb{C}) / \bigoplus_n M_{km_n}(\mathbb{C})$ is finite. This corona is canonically isomorphic to $M_k(A)$, so A is stably finite. Finally, let $p \leq q$ be equivalent projections in a stably finite algebra B , and choose a partial isometry w with $w^*w = q$ and $ww^* = p$. Then $w \in qBq$ is an isometry in the finite corner qBq , so $p = ww^* = q$. $\square$

Lemma 2.3 (one-sided order for the Kazhdan projection). *Let L have property (T), let B be a unital C^* -algebra, and let $\pi: L \rightarrow \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C_{\max}^*(L) \rightarrow B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\text{Fix } \pi(L)$. If ξ is L -fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B . $\square$

Theorem 2.4 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the conjugation actions are multiplicative modulo c_0 because the multiplicative defects of (V_n) tend to zero in operator norm. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2 . By the preceding estimate, $g \mapsto \tilde{\sigma}(g)$ is a homomorphism $G \rightarrow \mathcal{U}(\mathcal{B})$. So its restriction to L extends to a $*$ -homomorphism $C_{\max}^*(L) \rightarrow \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.3, and so $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.2 then gives $U^*PU = P$: U commutes with P , and both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence: a subsequence staying above some $\varepsilon > 0$ would lie in a free ultrafilter and give a nonzero ultralimit. Thus $\text{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. $\square$

Corollary 2.5. *Let $L \leq G$ have property (T), let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.4, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \rightarrow 0.$$

So

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \rightarrow 0 \quad (\ell \in L).$$

Intersecting over (V_n) proves the claim. $\square$

2.3. Normal Kazhdan subgroups and the MF radical.

Lemma 2.6 (central corona corners). *Let $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \text{rank}(q_n)$, unitary identifications $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n: G \rightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^)$ is the coordinate restriction of $q\rho(g)$, a unitary of the corner $q\mathcal{Q}_{\mathbf{d}}q$ with unit q .*

Proof. Lift q to projections q_n , and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \quad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero, and for $g, h \in G$,

$$\begin{aligned} & \|q_n U_n(gh) q_n - q_n U_n(g) q_n q_n U_n(h) q_n\| \\ & \leq \|U_n(gh) - U_n(g) U_n(h)\| + \|[q_n, U_n(h)]\| \longrightarrow 0, \end{aligned}$$

because $q_n U_n(g) U_n(h) q_n - q_n U_n(g) q_n U_n(h) q_n = q_n U_n(g) [U_n(h), q_n] q_n$ (compare [BDL, Proposition 2.4]). For each fixed g , let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\|\widetilde{W}_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel those coordinates. Put $r_n = \text{rank}(q_n)$, choose a unitary $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. $\square$

Theorem 2.7 (normal Kazhdan radical theorem). *Let G be countable and let $K \trianglelefteq G$ have property (T) with $K \leq R_{\infty \rightarrow 2}(G)$. Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K . Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ . The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It lies in $R_{\infty \rightarrow 2}(\Theta(G))$, because every operator norm asymptotic representation of $\Theta(G)$ pulls back along $G \rightarrow \Theta(G)$ to one of G .

Let $p_K \in C_{\max}^*(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K -fixed vectors. Normality of K gives

$$\Theta(g) p \Theta(g)^* = p \quad (g \in G).$$

The two ranges agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K , so every element of K has trivial corona image. Lemma 2.6 gives positive integers r_n and an operator

norm asymptotic representation $W_n: G \rightarrow \mathcal{U}(r_n)$. Let $\hat{\Theta}: G \rightarrow \mathcal{U}(\mathcal{Q}_r)$ be its corona homomorphism. Under the coordinatewise corner identifications, $\hat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_r$ faithfully. Since q commutes with $\Theta(K)$, the Kazhdan projection of K under $\hat{\Theta}$ is the coordinate restriction of $qp = 0$; so $\hat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\hat{\Theta}(s) - 1)^* (\hat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|} 1.$$

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$$

represent b . Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ then stays a positive Hilbert–Schmidt distance from I_{r_n} . This contradicts $s_0 \in K \leq R_{\infty \rightarrow 2}(G)$ for the replaced group G . Thus every corona homomorphism kills K . $\square$

Proof of Theorem A. For every $u \in \text{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.5 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

So normality of $R_{\infty \rightarrow 2}(G)$ gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.7 now gives $K \leq \text{Rad}_{\text{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\text{Rad}_{\text{MF}}(G) = G$. $\square$

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.4.

3. THE BINARY LEAVITT GROUP

The Leavitt relations give $\tau \in H$ with $\tau L \tau^{-1} \leq L$ for a copy L of $\text{EL}_3(R)$ and a nontrivial $d \in \mathfrak{D}_H(L)$. Simplicity of H then gives $\mathfrak{D}_H(L) = H$.

3.1. The self-compression. Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0 t_0, \quad q = s_1 t_1.$$

The relations (2) give

$$(4) \quad p + q = 1, \quad t_1 q s_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. Define a map $\Psi: M_3(R) \rightarrow M_3(R)$ by

$$(5) \quad \Psi(A) = qI_3 + s_0 A t_0.$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $q s_0 = t_0 q = 0$, and $t_0 s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0 I_3 t_0 = qI_3 + pI_3 = (p + q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0 \Psi(A) s_0 = A$, so Ψ is injective and restricts to an injective group homomorphism $\mathrm{GL}_3(R) \rightarrow \mathrm{GL}_3(R)$.

To realize Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $Y = X^{-1}$. Put

$$(6) \quad \tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R).$$

A direct block multiplication gives

$$(7) \quad \tau \mathrm{diag}(A, I_9) \tau^{-1} = \mathrm{diag}(\Psi(A), I_9).$$

Lemma 3.1. *The matrix τ defined in (6) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. For $B \in M_6(R)$, the block matrices $\begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$ and $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}$ lie in $\mathrm{EL}_{12}(R)$: they are the products of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, and no cross terms arise because distinct off-diagonal matrix units have product zero. Whitehead's identity

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$$

with $I = I_6$ then expresses τ as a product of six matrices of $\mathrm{EL}_{12}(R)$. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property (T), and $\tau \in H$ satisfies*

$$(8) \quad \tau L \tau^{-1} \leq L.$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. So the theorem of Ershov and Jaikin-Zapirain gives property (T) for $\text{EL}_{12}(R)$ and $\text{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L . These matrices generate L , so (8) follows. $\square$

3.2. Simplicity and the full MF radical.

Proposition 3.3. *The group $H = \text{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and so is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \text{EL}_{12}(R)$, the subgroup $N \leq \text{GL}_{12}(R)$ is normalized by $\text{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\text{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R))$. Fix $g \in N$. Since g centralizes $e_{ij}(1)$ for every $i \neq j$, comparison of entries in $gE_{ij} = E_{ij}g$ shows that $g = \lambda I_{12}$ for some $\lambda \in R^\times$. Centralizing $e_{ij}(a)$ now gives $\lambda a = a\lambda$ for every $a \in R$, and so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $g = 1$. Thus $N = 1$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 3.4. *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. So the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L , so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$(9) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ then gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since H is simple, the nontrivial element d normally generates H . $\square$

Proof of Theorem B. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. The identities $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ show inductively that Y generates all elementary matrices over R ; the commutator identity always has a third index because $12 \geq 3$. Thus $H = \langle Y \rangle$ is finitely generated, and Proposition 3.2 gives property (T).

The containment (8), the centrality of c relative to L , and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\text{Rad}_{\text{MF}}(H) = H$. Proposition 3.3 shows that H is nontrivial and simple.

The equality $\text{Rad}_{\text{MF}}(H) = H$ implies that every homomorphism from H to an MF group is trivial, and so that H is not MF.

$C_r^*(H)$ is separable, and its faithful canonical trace makes it stably finite. If an MF embedding $e: C_r^*(H) \rightarrow \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1-p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, contrary to the preceding conclusion. $\square$

4. A TORSION-FREE GROUP WITH FULL MF RADICAL

For an acylindrically hyperbolic group G , choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. We use Hull's small cancellation theorem [Hu, Theorem 7.1] as follows. Given a finite set $\Omega \subseteq G$, choose a ball in $\Gamma(G, A)$ containing Ω and apply Hull's injectivity-radius formulation. This gives the finite-set version below.

Theorem 4.1 (Hull's small cancellation theorem). *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A , and let $g_1, \dots, g_m \in G$ and a finite subset $\Omega \subseteq G$ be given. Then there is a surjective homomorphism $\varphi: G \rightarrow Q$ such that Q is acylindrically hyperbolic, $\varphi|_{\Omega}$ is injective, $\varphi(g_i) \in \varphi(N)$ for all i , and every element of finite order in Q is the image of an element of the same order in G .*

Hull's proof treats $m = 1$ by passing to $G/\langle\langle W \rangle\rangle_G$ for one word W and the general case by induction on m [Hu, proof of Theorem 7.1], so $\ker \varphi$ is the normal closure of m elements and Q is finitely presented when G is.

Lemma 4.2 (saturation). *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $\Omega \subseteq G$ be finite. Then there is a surjective homomorphism $\varphi: G \rightarrow Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi|_{\Omega}$ is injective, and $\varphi(N) = Q$.*

Proof. Because $N \trianglelefteq G$, one has $N \cap gNg^{-1} = N$ for every $g \in G$; this group is infinite because G is torsion-free and $N \neq 1$. Thus N acts non-elementarily on $\Gamma(G, A)$ by Osin [Os, Lemma 7.1]. Torsion-freeness makes N suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two

elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to N_0 with $g_1, \dots, g_m$ a finite generating set of G and with the set Ω . Then

$$Q = \langle \varphi(g_1), \dots, \varphi(g_m) \rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q,$$

so $Q = \varphi(N_0) = \langle \varphi(h_1), \varphi(h_2) \rangle$ and $\varphi(N) = Q$. The quotient is torsion-free by Theorem 4.1 and finitely presented because $\ker \varphi$ is normally generated by m elements. $\square$

Following Fournier-Facio [FF, §2], let U be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group [Chi]; let $F = F(a, b)$ be the free group of rank two; and let B_0 be a torsion-free hyperbolic group with property (T), obtained from the density model at a parameter between $1/3$ and $1/2$ [KK, Ol]. Small cancellation over the relatively hyperbolic pair $(U * B_0, U)$ gives a quotient Δ of $U * B_0$ in which U embeds and each generator of U equals an element of the image of B_0 ; so Δ is a finitely presented torsion-free quotient of B_0 [FF2, Proposition 2.3]; torsion-freeness is preserved by Osin's embedding theorem [OsSC, Theorem 2.4(5)]. Consequently, Δ has property (T). By universality, Δ contains a subgroup $\Delta_1 \times \Delta_2 \times F$ with $\Delta_i \cong \Delta$. Fix isomorphisms $\phi_i: \Delta \rightarrow \Delta_i$ and let E be the double HNN extension of Δ with stable letters u_1, u_2 and relations $u_i p u_i^{-1} = \phi_i(p)$ for $p \in \Delta$, so that $u_i \Delta u_i^{-1} = \Delta_i$. It is finitely presented, and torsion-free because an element of finite order fixes a vertex of the Bass–Serre tree and so is conjugate into Δ . The action of E on that tree is minimal and fixes no end, since Δ_1 and Δ_2 are proper subgroups of Δ [FF, §2], and the pointwise stabilizer of the vertices $u_1 \Delta$ and $u_2 \Delta$ is $\Delta_1 \cap \Delta_2 = 1$; so E is acylindrically hyperbolic by Minasyan–Osin [MO, Theorem 2.1]. Hull's common quotient theorem [Hu, Corollary 7.4], applied to the finitely generated torsion-free acylindrically hyperbolic groups E and B_0 , gives an acylindrically hyperbolic group G_0 with surjections $\pi: E \rightarrow G_0$ and $B_0 \rightarrow G_0$, where π is injective on a prescribed finite subset of E because E has no nontrivial finite normal subgroup. In the proof of that corollary, G_0 is obtained from the free product $E * B_0$ by two applications of Theorem 4.1, each with finitely many elements g_i ; so G_0 is torsion-free and, by the remark after Theorem 4.1, finitely presented. It has property (T) as a quotient of B_0 . Put $w = [a, b] \neq 1$ in F and require the prescribed finite subset to contain 1 and w . Then $\pi(w) \neq 1$.

Put $\Gamma = \pi(\Delta)$, $t = \pi(u_1)$, and $J = t^{-1} \pi(F) t$. Since F commutes with $\Delta_1 = u_1 \Delta u_1^{-1}$, the subgroup J centralizes Γ , and $t \Gamma t^{-1} = \pi(\Delta_1) \leq \Gamma$, so $t \in \text{Comp}_{G_0}(\Gamma)$. Moreover $t J t^{-1} = \pi(F) \leq \Gamma$, and Γ has property (T) as a quotient of Δ .

Lemma 4.3. *For every homomorphism $\rho: G_0 \rightarrow \bar{G}$,*

$$\rho(\pi([F, F])) \leq \mathfrak{D}_{\bar{G}}(\rho(\Gamma)).$$

Proof. Write $\bar{\Gamma} = \rho(\Gamma)$, $\bar{t} = \rho(t)$, and $\bar{J} = \rho(J)$. Then $\bar{t} \bar{\Gamma} \bar{t}^{-1} \leq \bar{\Gamma}$, $\bar{J}$ centralizes $\bar{\Gamma}$, and $\bar{t} \bar{J} \bar{t}^{-1} = \rho(\pi(F)) \leq \bar{\Gamma}$. For $c \in \bar{J}$ and $\ell \in \bar{\Gamma}$, the commutator

$[\bar{t}c\bar{t}^{-1}, \ell]$ lies in $\mathfrak{D}_{\bar{G}}(\bar{\Gamma})$ by (1). As c ranges over $\bar{J}$, the element $\bar{t}c\bar{t}^{-1}$ ranges over $\rho(\pi(F))$, and ℓ may be taken in $\rho(\pi(F)) \leq \bar{\Gamma}$. So the image of $[F, F]$ lies in $\mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. $\square$

Proof of Theorem C. Let N be the normal closure of $\pi([F, F])$ in G_0 . It is nontrivial because it contains $\pi(w)$. Lemma 4.2 applied to G_0 , N , and the finite set $\{1, \pi(w)\}$ gives $\varphi: G_0 \rightarrow Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi(N) = Q$, and $\varphi(\pi(w)) \neq 1$. The group Q has property (T) as a quotient of G_0 .

Let $r: Q \rightarrow \bar{G}$ be a surjection onto a nontrivial group and put $\rho = r \circ \varphi$. Since $\varphi(N) = Q$, the normal closure of $\rho(\pi([F, F]))$ in $\bar{G}$ is $\rho(N) = \bar{G}$. By Lemma 4.3, $\rho(\pi([F, F])) \leq \mathfrak{D}_{\bar{G}}(\rho(\Gamma))$, and since $\mathfrak{D}_{\bar{G}}(\rho(\Gamma))$ is normal in $\bar{G}$ and contains a normal generating set of $\bar{G}$, it equals $\bar{G}$. The group $\bar{G}$ has property (T) as a quotient of Q , and $\rho(\Gamma)$ has property (T) as a quotient of Γ . Theorem A, applied to $\bar{G}$ with the subgroup $\rho(\Gamma)$ and $K = \bar{G}$, gives $\text{Rad}_{\text{MF}}(\bar{G}) = \bar{G}$. Taking r the identity gives $\text{Rad}_{\text{MF}}(Q) = Q$, and a nontrivial group equal to its own MF radical is not MF. $\square$

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